Conditional Probability

Probability Worksheet · Grade 10–12

Name: ___________________________________ Date: ____________________

Learning Objectives

- Apply the multiplication rule for dependent events to find the probability of two events both occurring
- Use the conditional probability formula P(B|A) = P(A and B) / P(A) when probabilities are given
- Distinguish between independent and dependent events and choose the correct probability method

Problems

1. A bag contains 5 red marbles and 3 blue marbles. You draw one marble and do NOT replace it. What is the probability that the first marble is red?

$$P(A) = \frac{5}{8}$$

2. A standard deck has 52 cards. You draw one card and do NOT replace it. Event A is drawing a heart. Given that event A occurred, what is the probability that the second card drawn is also a heart (event B given A)?

$$P(B \mid A) = \frac{12}{51}$$

3. Using the multiplication rule for dependent events, find the probability of drawing two aces in a row from a standard 52-card deck without replacement.

$$P(A \cap B) = \frac{4}{52} \times \frac{3}{51}$$

4. A bag has 6 green and 4 yellow marbles. Two marbles are drawn without replacement. Use the multiplication rule for dependent events to find the probability that both marbles are green.

$$P(A \cap B) = \frac{6}{10} \times \frac{5}{9}$$

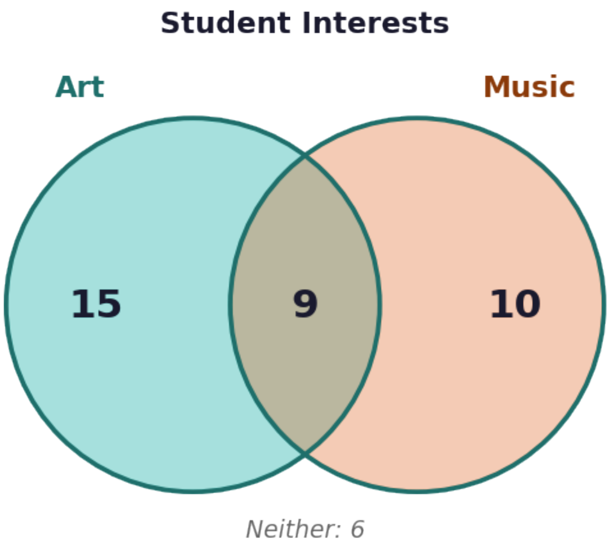
5. In a class, 60% of students like math. Of those who like math, 40% also like science. Using the multiplication rule for dependent events, what percent of students like both math and science?

$$P(A \cap B) = 0.60 \times 0.40$$

6. Use the Type 2 conditional probability formula. In a survey, 80% of people exercise regularly. 48% exercise regularly AND eat healthy. What percent of those who exercise regularly also eat healthy?

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.48}{0.80}$$

7. A school survey shows the following. Use the Venn diagram to answer: What is the probability that a randomly selected student who likes art also likes music?



8. From a standard deck of 52 cards, three cards are drawn without replacement. Use the multiplication rule for dependent events to find the probability that all three cards are kings. Round to 4 decimal places.

$$P = \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50}$$

9. Use the table below and the Type 2 conditional probability formula to find the probability that a student chosen at random passed the final exam, given that they completed all homework assignments.

	Passed Exam	Did Not Pass Exam	Total
Completed Homework	45	5	50

	Passed Exam	Did Not Pass Exam	Total
Did Not Complete Homework	10	40	50
Total	55	45	100

10. In a survey of 200 students, 110 play a sport, 90 play an instrument, and 45 play both a sport and an instrument. A student is selected at random. Find: (a) the probability that the student plays an instrument given that they play a sport, and (b) the probability that the student plays a sport given that they play an instrument. Are these two conditional probabilities equal? Explain.

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \quad P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability — Answer Key

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Answer Key

1. Answer: 5/8 or 0.625
- There are 5 red marbles out of 8 total marbles.
- P(red) = 5/8 = 0.625

2. Answer: 12/51 ≈ 0.235
- After drawing one heart, 12 hearts remain out of 51 total cards.
- P(B|A) = 12/51 ≈ 0.235

3. Answer: 12/2652 ≈ 0.45%
- P(first ace) = 4/52
- P(second ace | first ace) = 3/51
- P(A and B) = (4/52) × (3/51) = 12/2652 ≈ 0.0045 or about 0.45%

4. Answer: 30/90 = 1/3 ≈ 33.3%
- P(first green) = 6/10
- P(second green | first green) = 5/9
- P(both green) = (6/10) × (5/9) = 30/90 = 1/3 ≈ 33.3%

5. Answer: 24%
- P(math) = 0.60
- P(science | math) = 0.40
- P(math and science) = 0.60 × 0.40 = 0.24 = 24%

6. Answer: 60%
- P(A) = 0.80 (exercise regularly)
- P(A and B) = 0.48 (exercise and eat healthy)
- P(B|A) = 0.48 / 0.80 = 0.60 = 60%

7. Answer: 9/24 = 3/8 = 37.5%
- P(A) = students who like Art = 15 + 9 = 24
- P(A and B) = students who like both = 9
- P(B|A) = 9/24 = 3/8 = 0.375 = 37.5%

8. Answer: 24/132600 ≈ 0.0002 or 0.018%
- P(1st king) = 4/52
- P(2nd king | 1st king) = 3/51
- P(3rd king | first two kings) = 2/50
- P(all three kings) = (4/52) × (3/51) × (2/50) = 24/132600 ≈ 0.000181 ≈ 0.018%

9. Answer: 45/50 = 0.90 = 90%

- Event A = completed all homework, P(A) = 50/100 = 0.50
- Event A and B = completed homework AND passed exam, P(A and B) = 45/100 = 0.45
- P(B|A) = P(A and B) / P(A) = 0.45 / 0.50 = 0.90 = 90%

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10. Answer: P(instrument | sport) = 45/110 ≈ 40.9%; P(sport | instrument) = 45/90 = 50%; They are NOT equal.

- P(A) = P(sport) = 110/200 = 0.55
- P(B) = P(instrument) = 90/200 = 0.45
- P(A and B) = 45/200 = 0.225
- P(B|A) = 0.225 / 0.55 = 45/110 ≈ 0.409 = 40.9%
- P(A|B) = 0.225 / 0.45 = 45/90 = 0.50 = 50%
- The two conditional probabilities are NOT equal because the size of the conditioning event (sport vs instrument) differs.
